Charan H S

Computer Science (Data Science) Undergraduate | AI/ML Developer | Research Enthusiast

Email: charanhs359@gmail.com **Phone:** +91 8296536615 **LinkedIn:** linkedin.com/in/charan-hs **GitHub:** github.com/Charan359

EDUCATION

Bachelor of Engineering (Data Science)

Nagarjuna College of Engineering and Technology

2024 - 2028 (Expected) **8.41 CGPA (4th Sem)**

12th – CBSE

Sri Chaitanya Techno School, Karnataka — 80.8%

PERSONAL PROJECTS

URBANPULSE AI — Smart City Intelligence Platform

(Team Lead | 13 Core Modules)

- Spearheaded smart city platform featuring 6-way route optimization evaluating 14 multi-factor dimensions including thermal comfort, AQI levels, pedestrian safety, and accessibility.
- Engineered multilingual Voice AI (15 languages) with voice-first talkback navigation for visually impaired pedestrians, sensor fusion dashboards, and live Leaflet overlays without external API keys.

Live Link : https://urbanpulse-ai.netlify.app/

React 19, TypeScript, Tailwind CSS, Supabase, Groq (Llama 3.1), Leaflet, Web Speech API.

WORLDMONITOR — Cybersecurity Assessment Platform

(SIH 2026 Nominated Solution)

- Built full-stack vulnerability assessment platform comprising 7 specialized security engines inspecting authentication, authorization (BOLA/IDOR), API endpoints, and database privacy controls.
- Engineered hybrid SAST + DAST AI correlation using Semgrep AST, differential response analysis, cryptographic SHA-256 evidence integrity hashing, and automated patch re-testing.

Live Link : https://securix-world-monitor.vercel.app/

FastAPI, Python, Semgrep AST, React, WebSockets, SHA-256 Verification, REST APIs.

DVR/NVR FORENSIC ANALYZER — Video Recovery & Forensics

(SIH 2026 Problem Statement: SIH26150)

- Built interactive digital forensics suite simulating physical drive scanning, software write-blocking, bit-by-bit imaging, and proprietary CCTV file system parsing (DHFS, WFS, RSFS).
- Implemented automated video carving algorithms to extract fragmented H.264/H.265 streams from unallocated disk space, timeline reconstruction, and immutable SHA-256 chain-of-custody logs.

Live Link : https://dvr-nvr-forensic-analyzer.vercel.app/

JavaScript, HTML5, CSS3, Forensic Data Structures, SHA-256 Hashing, ISO/IEC 27037.

GROCERY DELIVERY ANALYTICS — Power BI Dashboard

- Developed interactive retail analytics dashboard analyzing Rs. 12.01 Lakhs in cumulative sales across 1,559 items to optimize inventory distribution, outlet performance, and logistics efficiency.
- Built comprehensive DAX data models and visual KPI cards tracking outlet sizes, location tiers (Tier 1-3), item sales categories, and average ratings to extract actionable operational insights.

Repository : https://github.com/Charan359/Grocery-Delivery-Analytics

Power BI, Microsoft Excel, Data Analytics, DAX Formulas, Data Modeling & Visualization.

RESEARCH & PUBLICATIONS

PrivAgentShield: Policy-Aware Runtime Framework for Multi-Agent LLMs

First Author | Accepted for Springer Nature Proceedings

ICMMDU 2026 Conference (Paper ID: 190)

- Formulated non-compensatory minimax Leakage Risk Index (LRI*) unifying payload sensitivity, lattice clearance dominance, and absorbing Markov reachability for multi-agent LLM systems.
- Achieved 65.0% privacy leakage prevention, 74.2% task completion preservation, and 1.66 ms mean runtime latency across 31 empirical benchmark execution traces without model retraining.

Live Demo : https://priv-agent-shield-bice.vercel.app/

SKILLS

Python | Java | C | C++ | SQL | Machine Learning | Data Analytics | Power BI | Pandas | NumPy | Scikit-learn | Matplotlib | React | HTML | CSS | JavaScript | Flask | MySQL | MongoDB | Git | GitHub | Excel | Jupyter Notebook | DSA | OOP

ACHIEVEMENTS

Coordinator — Vibexathon 1.0 (State Hackathon)

Dept. of CSE (Data Science) & IEEE NCET | 16-17 March 2026

Recognized with official Certificate of Appreciation for leadership and end-to-end event management of a 24-hour State-Level Hackathon.

Nominated — Smart India Hackathon 2026

Ministry of Education & NCET Innovation Cell

Qualified internal college selection rounds and nominated for national submission across two problem tracks: Web Security Scanner and Forensic Video Carving.

First Author — International Research Acceptance

ICMMDU 2026 Conference | Springer Nature

Original multi-agent LLM security paper (PrivAgentShield) accepted for publication in Springer Nature conference proceedings among top peer-reviewed entries.

CERTIFICATES

Oracle Cloud Infrastructure 2025 Certified AI Foundations
Validated core competence in AI architectures, ML workloads, and OCI infrastructure services.

30 Days MasterClass in Data Analytics — NoviTech R&D
Hands-on industry training in exploratory data analysis, business intelligence, and visual analytics.

Applied Database Systems 23ai — Oracle Academy
Demonstrated proficiency in modern AI-integrated database design, schemas, and advanced SQL.

Python Essentials 1 — Cisco Networking Academy
Mastery of core Python scripting, data structures, algorithm design, and procedural development.

Software Engineering & ALM / Agile & DevOps — Infosys
Enterprise training in software development lifecycle, Agile practices, CI/CD, and DevOps workflows.

Java Fundamentals — Oracle Academy
Foundational expertise in object-oriented programming, classes, inheritance, and core Java libraries.

LANGUAGES

English
Full Professional Proficiency

Kannada
Native or Bilingual Proficiency

Hindi & Telugu
Conversational Proficiency

INTERESTS

Exploring AI & Emerging Tech | Research & Innovation | Problem Solving | Hackathons & Ideathons | Event Coordination & Teamwork | Data Analytics & DBMS